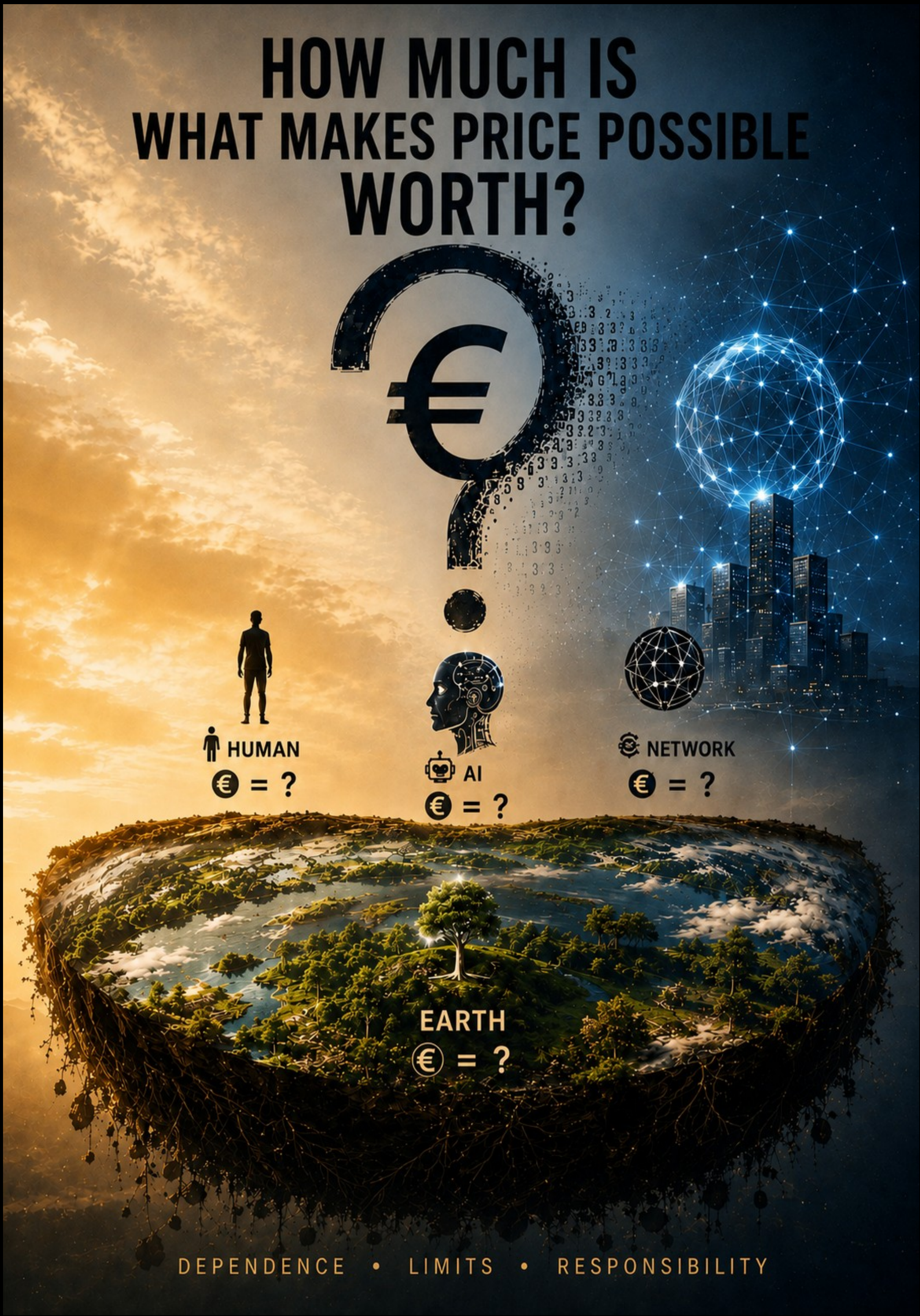

HOW MUCH IS
WHAT MAKES PRICE POSSIBLE
WORTH?
HUMAN
€ = ?
AI
€ = ?
NETWORK
€ = ?
EARTH
€ = ?
DEPENDENCE • LIMITS • RESPONSIBILITY

PRICE IS NOT THE BASE

We can put a price on many things:

🌲 wood
💧 water
🌾 production
⚡ energy
🏭 infrastructure
🌐 services

But that does not mean we can assign an equivalent price to what makes those things possible.

The question changes

Not only:

HOW MUCH DOES IT COST?

but:

WHAT DOES ITS ABILITY TO HAVE A PRICE DEPEND ON?

Central distinction

💰 = measure within the system

but:

🌍 = material condition of the system

Therefore:

💰 ∈ 🌐

and:

🌐 depends on 🦧 🌍

Formula

partial price ≠ total value
economic value ≠ condition of possibility

Central statement

WHAT MAKES PRICE POSSIBLE
IS NOT NECESSARILY EQUIVALENT TO A PRICE.

DEPENDENCE ≠ OWNERSHIP

A company may not own what it depends on.

It may not have:

🌲 forest

on its balance sheet.

But it may depend on:

💧 water regulated by that forest

It may not own:

🌍 territory

but it may depend on:

soil · climate · water · biodiversity · energy · stability

The accounting problem

What does not appear as property may still be:

condition of production
condition of profitability
condition of continuity

Therefore:

not being on the balance sheet

does not mean:

not sustaining the balance sheet

When the base degrades

➖ 🌍

it can become:

➖ production
➕ costs
➖ profitability
➕ risk

and then:

➖ 🙆 ➖ 🌐

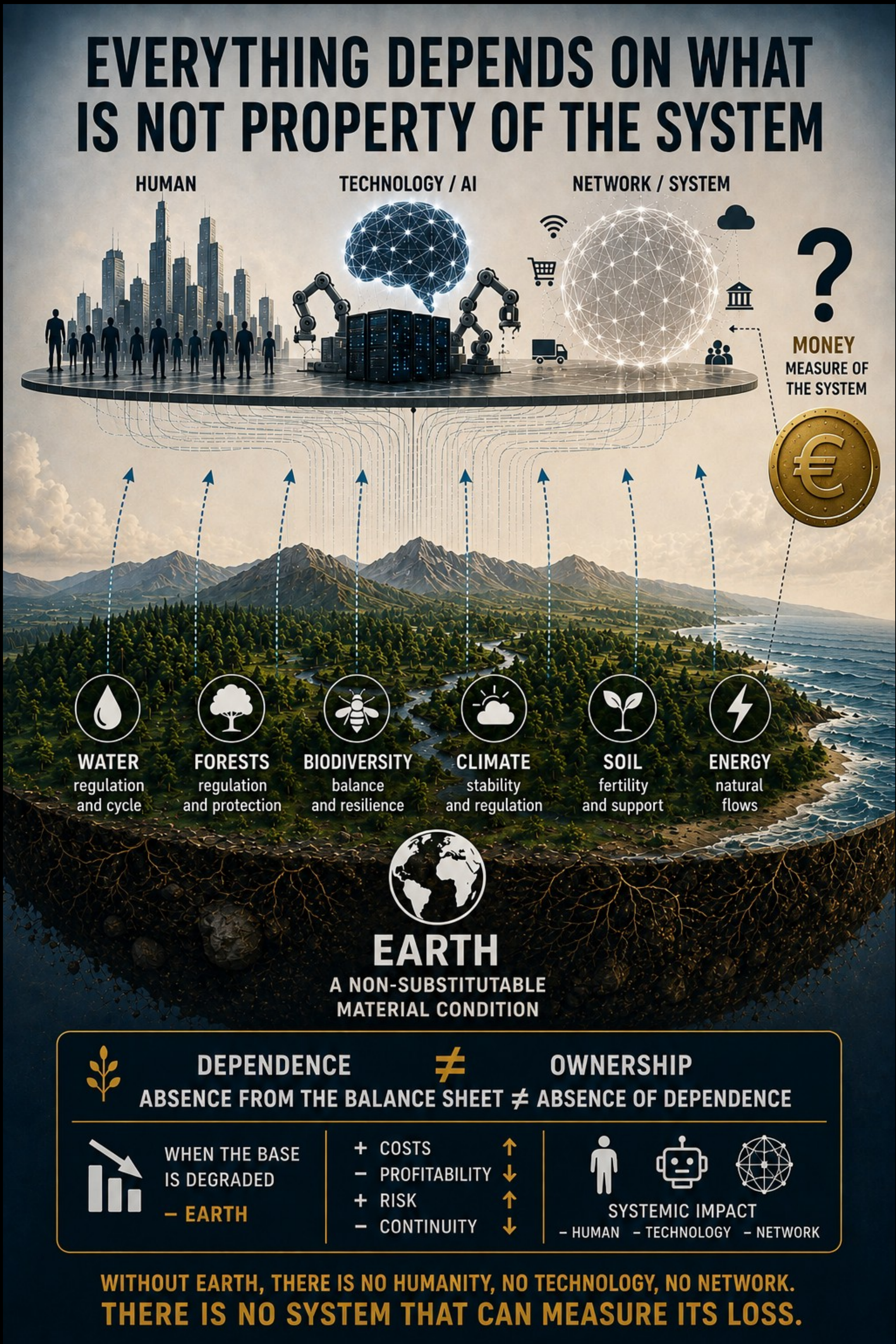
Formula

material dependence ≠ accounting ownership

absence from the balance sheet ≠ absence of dependence

Central statement

WHAT A COMPANY DOES NOT OWN
MAY BE EXACTLY WHAT IT DEPENDS ON.


EVERYTHING DEPENDS ON WHAT IS NOT PROPERTY OF THE SYSTEM
HUMAN
TECHNOLOGY / AI
NETWORK / SYSTEM
?
MONEY
MEASURE OF THE SYSTEM
€
WATER
regulation and cycle
FORESTS
regulation and protection
BIODIVERSITY
balance and resilience
CLIMATE
stability and regulation
SOIL
fertility and support
ENERGY
natural flows
EARTH
A NON-SUBSTITUTABLE MATERIAL CONDITION
DEPENDENCE ≠ OWNERSHIP
ABSENCE FROM THE BALANCE SHEET ≠ ABSENCE OF DEPENDENCE
WHEN THE BASE IS DEGRADED
– EARTH
+ COSTS ↑
– PROFITABILITY ↓
+ RISK ↑
– CONTINUITY ↓
SYSTEMIC IMPACT
– HUMAN – TECHNOLOGY – NETWORK
WITHOUT EARTH, THERE IS NO HUMANITY, NO TECHNOLOGY, NO NETWORK.
THERE IS NO SYSTEM THAT CAN MEASURE ITS LOSS.

WHEN DEPENDENCE BECOMES COST

While the base functions, much of its contribution remains invisible.

🌍 → 💧 🌲 🌾 🌡️ → 👷 🏭 🌐

The system receives those material conditions and treats them as context.

But when they begin to degrade:

➖ 💧
➖ 🌲
➖ 🌾
➕ 🌡️
➖ 🌍

what seemed external reappears inside the system as:

➕ 💰 costs
➖ 📈 productivity
➕ ⚠️ risk
➖ 🏭 continuity
➖ 👷 human margin

The paradox

First:

the base does not appear on the balance sheet.

Then:

its deterioration appears everywhere on the balance sheet.

Scarce water has a price.
Cooling has a price.
Lost production has a price.
Repair has a price.
Adaptation has a price.

Formula

➖ 🌍 → ➕ 💰 → ➖ 👷 → ➖ 🌐

Central statement

PRICE CAN MEASURE PART OF THE DAMAGE.
NOT NECESSARILY THE VALUE OF WHAT HAS BEEN LOST.

PRICE COMES AFTER

The economic cost usually becomes visible when material dependence has already begun to deteriorate.

First:

🌍 → sustains

Then:

➖🌍 → cost appears

Then the calculations begin:

➖💧 → 💰 scarcity
➖🌾 → 💰 loss
➖🏭 → 💰 disruption
🥵 → 💰 adaptation
🔥 → 💰 repair

The temporal problem

Price can register a consequence.

But it does not automatically restore:

💧
🌲
🌾
🌍

Therefore:

damage → valuation → compensation

does not equal:

damage → restoration

Formula

➖🌍 + ⏳ → ➕💰

but:

➕💰 ≠ ➕🌍

Central distinction

compensating ≠ restoring
paying ≠ giving back
valuing ≠ recovering

Central statement

PUTTING A PRICE ON DAMAGE DOES NOT REVERSE THE DAMAGE. COMPENSATING FOR A LOSS DOES NOT MEAN RECOVERING THE CONDITION THAT WAS LOST.

MONEY MEASURES FROM WITHIN

Money does not observe the system from outside.

It is part of it.

💰 ∈ 🌐

and:

🌐 depends on 🦧 🌍

The limit

As long as there are:

🦧
🤖
🌐

there can be:

markets
prices
balance sheets
credit
insurance
valuations

But if degradation reaches the conditions that sustain those same systems:

➖🌍 → ➖🦧 → ➖🤖 → ➖🌐

the capacity to:

💰 → measure

also decreases.

The paradox

The system asks:

🌍 💰 ❓

But that question depends on the continued existence of:

🦧 + 🌐 + institutions + markets

Formula

💰 measures from within

not:

💰 measures from outside

Central statement

MONEY CAN MEASURE LOSSES WITHIN THE SYSTEM.
IT CANNOT BECOME AN EXTERNAL MEASURE OF THE SYSTEM IT DEPENDS ON.

WHAT CANNOT BE SUBSTITUTED

An economic valuation can work when some possibility of substitution exists:

resource A → cost → resource B
damaged infrastructure → repair
lost production → compensation

But some conditions have no equivalent functional substitute.

Limit examples

🌍 💰 ❓
🌞 💰 ❓
🌑 💰 ❓

The question is not asking for a number.

It is meant to show the limit of the very operation of assigning a price.

Because:

🌍 is not just another asset within 🌐.

It is the material base on which depend:

🦧
🤖
🌐

and also:

💰

Distinction

having a price ≠ being substitutable
economic value ≠ functional equivalent

Formula

no functional substitute → total price insufficient

Central statement

WHAT HAS NO FUNCTIONAL SUBSTITUTE
CANNOT BE REDUCED TO AN EQUIVALENT PRICE.

WHEN THE QUESTION EXPANDS

The initial question was:

🌍 💰 ❓

But following the chain of dependence reveals something broader:

🦧 💰 ❓
🤖 💰 ❓
🌐 💰 ❓

Because the economic system does not value from outside.

It values from within what may also lose continuity.

The paradox

The system can calculate:

cost of losing water
cost of losing soil
cost of losing biodiversity
cost of losing production

But if accumulated loss compromises:

🦧
🤖
🌐

then it also compromises the system itself that is capable of continuing to calculate.

Formula

🌍 → 🦧 → 🤖 → 🌐 → 💰

Therefore:

💰 does not precede dependence
💰 depends on dependence

Limit question

WHAT DOES IT MEAN TO PUT A PRICE ON A LOSS THAT MAY END UP AFFECTING THE VERY SYSTEM THAT ASSIGNS THE PRICE?

Central statement

**THE SYSTEM CAN VALUE ITS LOSSES.
IT CANNOT PLACE ITSELF OUTSIDE THEM.**

WHO PAYS FOR THE LOSS?

When a common condition loses margin:

➖🌍

the impact does not disappear.

It is distributed.

It may reappear as:

➕💰 water
➕💰 cooling
➕💰 health
➕💰 repair
➕💰 adaptation

But each 1 🦧 has a different capacity to absorb it.

SAME LOSS · DIFFERENT IMPACT

🦧 with margin → buys adaptation
🦧 with little margin → loses economic margin
🦧 with no margin → receives the impact directly

The paradox

A common loss can become millions of individual responses:

➖🌍 → 💰🦧 $_1$ + 💰🦧 $_2$ + 💰🦧 $_3$ + …

but:

➕ individual adaptation ≠ ➕🌍

The cost can be privatized while the common condition continues to deteriorate.

Formula

➖🌍 → ➕💰🦧

but:

➕💰🦧 ≠ recovery of 🌍

Central statement

THE LOSS IS COMMON.
THE CAPACITY TO PAY FOR IT IS NOT.
PRIVATIZING THE COST DOES NOT RECOVER THE COMMON CONDITION.

MEASURING MORE DOES NOT CHANGE THE DIRECTION

We can increase the precision of calculation:

➕ data
➕ indicators
➕ metrics
➕ models
➕ valuation

But if the reference for decision-making remains the same:

same 🧭

then:

➕ measurement ≠ correction

The paradox

The system can know with increasing precision:

what it is losing
how much it costs
which sectors it affects
which risks it accumulates

and, at the same time, continue:

➕ 🌐 ➕ 🤖 ➕ 🚀 ➕ ⏳

while:

➖ 🌍 ➖ 🦧

The problem is not lack of information

Information only changes the trajectory when it enters:

decision
priority
rule
limit
return

Formula

➕ knowledge ≠ reorientation
➕ precision ≠ continuity

and:

➕ 🌐 ➕ 🤖 ➕ 🚀 ➕ ⏳ ➖ 🧭
🆚
➖ 🌍 ➖ 🦧

Central statement

MEASURING DETERIORATION MORE PRECISELY DOES NOT STOP IT.
EVIDENCE CHANGES THE DIRECTION ONLY WHEN IT CHANGES THE DECISION.

PRINCIPLE OF NON-EQUIVALENCE

From everything above, a minimal rule can be derived:

AN ECONOMIC REPRESENTATION IS NOT EQUIVALENT TO THE MATERIAL CONDITION IT REPRESENTS.

Because:

💰 can:

measure
compare
record
compensate
signal risk

But it cannot guarantee:

reversibility
substitution
regeneration
continuity

Central distinction

price = representation
territory = condition

Therefore:

representation ≠ condition

and:

💰 ≠ 🌍

Not because:

🌍 = without value

but because:

🌍 ≠ a complete functional equivalent in 💰

The limit

➖🌍 → 💰 valuation

can produce:

cost
compensation
accounting

but none of them demonstrates by itself:

➕🌍

Central formula

VALUING ≠ SUBSTITUTING
COMPENSATING ≠ RESTORING
PAYING ≠ GIVING BACK

Central statement

PRICE CAN REPRESENT A LOSS.
IT CANNOT BECOME THE EQUIVALENT OF WHAT THAT LOSS REPRESENTS.

SCOPE NOTE

This proposal does not claim that:

💰 = useless

nor that:

economic valuation = error

Valuation can help to:

identify dependencies
make costs visible
compare scenarios
anticipate risks
guide decisions

The limit appears when 💰 stops functioning as a partial tool and begins to be interpreted as equivalent to what sustains the system.

Distinction

price → information
dependence → material relation
continuity → condition

Therefore:

price ≠ dependence
price ≠ substitutability
price ≠ continuity

We are not saying

DO NOT value

We are saying:

VALUE WITHOUT CONFUSING REPRESENTATION WITH EQUIVALENCE.

Criterion

💰 → can enter the decision

but:

🌍 → must continue to enter verification

Because a figure can change.

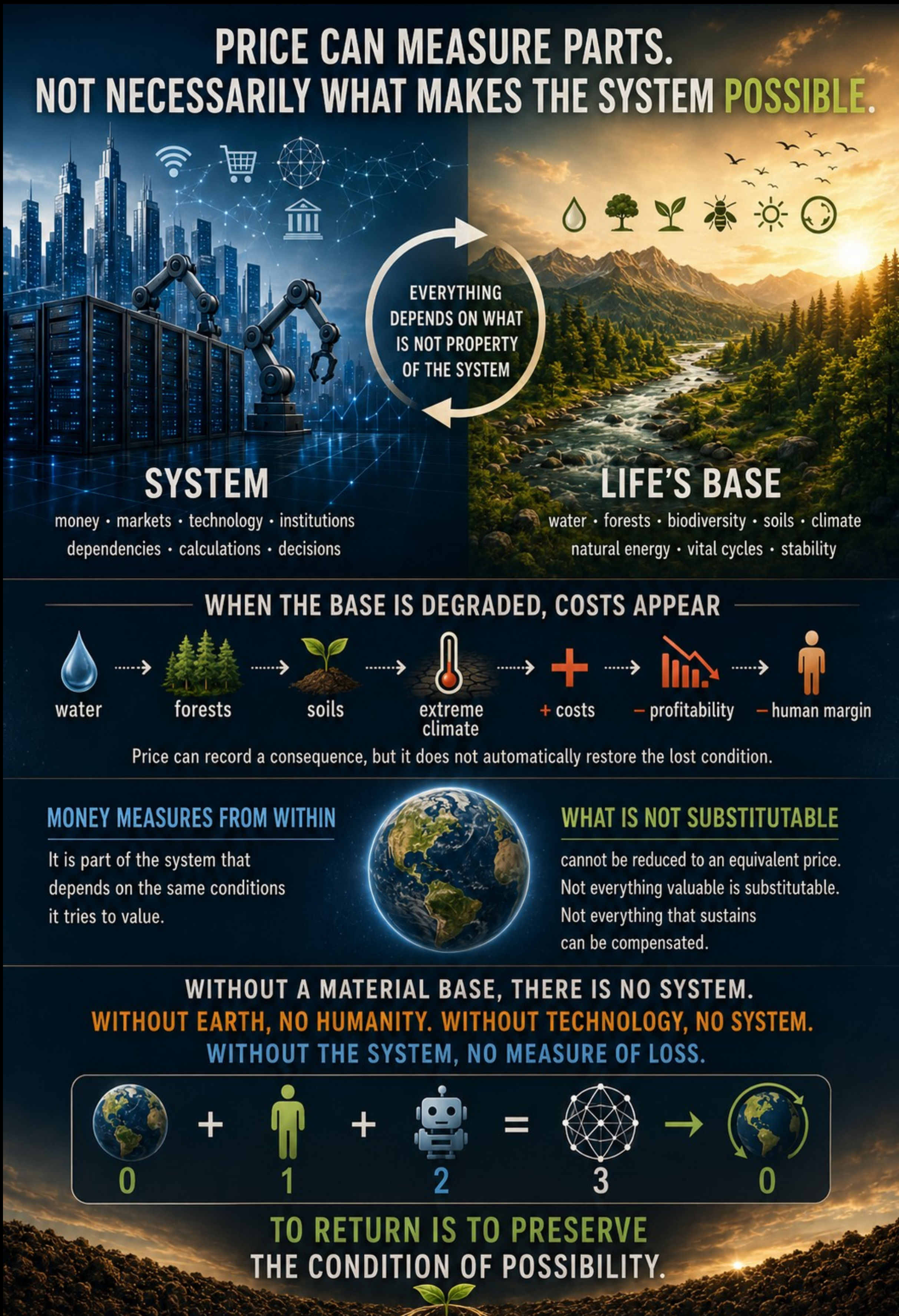
A material condition that has been lost may not change with the same ease or within the same timeframe.

Central statement

THE PROBLEM IS NOT CALCULATION.
IT IS FORGETTING WHAT REMAINS OUTSIDE THE CALCULATION.

HOW MUCH IS WHAT MAKES PRICE POSSIBLE WORTH?

NOT EVERYTHING THAT CAN BE VALUED CAN BE SUBSTITUTED.


PRICE CAN MEASURE PARTS.
NOT NECESSARILY WHAT MAKES THE SYSTEM POSSIBLE.
EVERYTHING DEPENDS ON WHAT IS NOT PROPERTY OF THE SYSTEM
SYSTEM
money · markets · technology · institutions
dependencies · calculations · decisions
LIFE'S BASE
water · forests · biodiversity · soils · climate
natural energy · vital cycles · stability
WHEN THE BASE IS DEGRADED, COSTS APPEAR
water
forests
soils
extreme climate
+ costs
– profitability
– human margin
Price can record a consequence, but it does not automatically restore the lost condition.
MONEY MEASURES FROM WITHIN
It is part of the system that depends on the same conditions it tries to value.
WHAT IS NOT SUBSTITUTABLE
cannot be reduced to an equivalent price.
Not everything valuable is substitutable.
Not everything that sustains can be compensated.
WITHOUT A MATERIAL BASE, THERE IS NO SYSTEM.
WITHOUT EARTH, NO HUMANITY. WITHOUT TECHNOLOGY, NO SYSTEM.
WITHOUT THE SYSTEM, NO MEASURE OF LOSS.
+
+
=
0
1
2
3
0
TO RETURN IS TO PRESERVE
THE CONDITION OF POSSIBILITY.

CLOSING · BEFORE PRICE

The initial question seemed economic:

HOW MUCH IS WHAT MAKES PRICE POSSIBLE WORTH?

After following the argument, the answer can no longer be a number.

Because:

🌍0 → 🦧1 → 🤖2 → 🌐3 → 💰

describes a sequence of dependence, not a scale of prices.

Therefore:

🌍 💰 ❓

does not ask:

WHAT NUMBER SHOULD WE ASSIGN?

It asks:

CAN A CONDITION OF POSSIBILITY HAVE A TOTAL EQUIVALENT WITHIN THE SYSTEM IT MAKES POSSIBLE?

Final distinction

💰 → can measure
💰 → can signal
💰 → can compensate

but:

💰 ≠ condition

Closing formula

🌍 = material base
🦧 = responsibility
🤖 = amplification
🌐 = connection
💰 = tool

Final statement

BEFORE PRICE COMES DEPENDENCE.
BEFORE PROFITABILITY COMES CONTINUITY.
NOT EVERYTHING THAT CAN BE VALUED CAN BE SUBSTITUTED.

🌍0 → 🦧1 → 🤖2 → 🌐3 → 💰

PRICE IS INSIDE THE SYSTEM.
THE MATERIAL BASE MAKES THE SYSTEM POSSIBLE.

LEGEND

🌍 0
Territory / material base

🦺 1
Human / responsibility

🤖 2
AI / technical amplification

🌐 3
Network / created architecture

💰
Price / economic valuation

➕ / ➖
Increase / decrease

⏳
Time / accumulation

🚀
Acceleration / expansion

🧭
Direction / course

💥
Possible rupture or critical loss

∈
Belongs to / forms part of

→
Dependence / sequence / return, depending on context

Reference formula

🌍 0 → 🦺 1 → 🤖 2 → 🌐 3 → 💰

material base → human responsibility → technical amplification → connection → economic valuation

Central distinction

🌍 = condition
💰 = tool

Closing

PRICE IS INSIDE THE SYSTEM.
THE MATERIAL BASE MAKES THE SYSTEM POSSIBLE.

HOW MUCH IS WHAT MAKES PRICE POSSIBLE WORTH?

Robles Rionegro, D. A. (2026). *How much is what makes price possible worth?* (Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.22148748